

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

AI Problem Solver Portfolio (CO1)

Assignment No: 1

AI-BASED DISASTER RELIEF SUPPLY DRONE

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Selected Problem: AI-Based Disaster Relief Supply Drone

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1.PROBLEM DESCRIPTION

An **AI-Based Disaster Relief Supply Drone** is an intelligent autonomous aerial system developed to deliver essential emergency supplies such as medicines, food packets, drinking water, first-aid kits, and communication devices to disaster-affected areas without direct human intervention. During natural disasters such as floods, earthquakes, cyclones, and landslides, damaged roads and blocked transportation routes often delay rescue operations. The drone utilizes **Artificial Intelligence (AI)** to analyze the environment, determine the safest and shortest flight path, avoid obstacles, and navigate efficiently to the destination. By integrating GPS, onboard sensors, cameras, and autonomous navigation techniques, the drone continuously monitors its surroundings and dynamically replans its route whenever unexpected obstacles or hazards are detected. This intelligent system improves emergency response time, reduces operational risks, and ensures timely delivery of life-saving supplies to affected communities.

Problem Summary

Aspect	Description
System Name	AI-Based Disaster Relief Supply Drone
Primary Objective	Deliver emergency supplies safely and quickly to disaster-affected areas
Application Areas	Floods, Earthquakes, Cyclones, Landslides, Wildfires
AI Function	Route planning, obstacle avoidance, autonomous navigation, decision-making
Essential Supplies	Medicines, Food, Water, First-aid Kits, Communication Devices
Key Benefits	Reduced response time, improved safety, efficient disaster management, minimal human intervention

3. PEAS Description

The **PEAS (Performance Measure, Environment, Actuators, and Sensors)** framework is used to define the operational characteristics of the **AI-Based Disaster Relief Supply Drone**. It provides a structured representation of how the intelligent drone perceives its surroundings, makes autonomous decisions, and performs actions to accomplish its mission. The **Performance Measure** focuses on delivering emergency supplies quickly, safely, and accurately while minimizing flight time, battery consumption, and navigation errors. The **Environment** includes dynamic disaster-affected areas such as flood zones, earthquake-hit regions, forests, relief camps, hospitals, and temporary shelters, where unexpected obstacles and changing weather conditions may occur. The drone performs its tasks using **Actuators** such as brushless motors, propellers, navigation controllers, cargo release mechanisms, and communication modules that enable flight control and package delivery.

3. Type of Intelligent Agent with Justification

Agent Type: Goal-Based Agent

The **AI-Based Disaster Relief Supply Drone** is classified as a **Goal-Based Intelligent Agent** because its primary objective is to deliver emergency supplies from the relief center to disaster-affected locations safely and efficiently. The drone continuously monitors its surroundings, determines its current position, and plans an optimal flight path toward the destination using AI search algorithms such as **Breadth-First Search (BFS)** or **A***. Unlike a simple reflex agent that responds only to immediate sensor inputs, a goal-based agent evaluates different possible routes and selects the one that best satisfies its delivery objective.

During flight, the drone can adapt to dynamic environmental changes such as damaged structures, restricted airspace, adverse weather conditions, or unexpected obstacles by recalculating an alternative path. This ability to plan, make decisions based on future states, and modify its actions according to real-time information makes the **Goal-Based Agent** the most suitable model for autonomous disaster relief operations. As a result, the drone ensures reliable, safe, and timely delivery of essential supplies while minimizing risks and improving emergency response efficiency.

4. Environment Classification

The operating environment of the **AI-Based Disaster Relief Supply Drone** can be classified based on the characteristics of the disaster response scenario. The drone must perceive its surroundings, make intelligent decisions, and adapt to changing environmental conditions while completing its delivery mission.

Property	Classification	Explanation
Observable	Partially Observable	The drone receives information from cameras, GPS, and onboard sensors, but factors such as smoke, rain, fog, or damaged infrastructure may temporarily reduce visibility.
Agents	Multi-Agent	The environment consists of rescue teams, helicopters, emergency vehicles, civilians, and other drones operating simultaneously.
Static / Dynamic	Dynamic	The disaster environment changes continuously due to weather conditions, moving obstacles, rescue operations, and unexpected emergencies.
Discrete / Continuous	Continuous	The drone flies in continuous three-dimensional space where position, altitude, speed, and direction change continuously.
Sequential / Episodic	Sequential	Every navigation decision influences the next movement, battery usage, and the overall success of the mission.
Known / Unknown	Partially Known	The drone has access to maps and GPS data before take-off, but real-time obstacles and environmental changes are discovered only during flight.

5. Problem Formulation

Initial State

The drone starts from the **Disaster Relief Control Center** carrying emergency supplies to be delivered to a disaster-affected relief camp.

Goal State

The drone successfully reaches the **Emergency Shelter** and safely delivers the emergency supplies.

Operators (Actions)

- Take Off
- Move Forward
- Turn Left
- Turn Right
- Ascend / Descend
- Avoid Obstacle
- Deliver Supplies
- Return to Base

State Space

The state space consists of all possible locations the drone can visit during the mission, including:

- Disaster Relief Control Center
- Checkpoint A
- Checkpoint B
- Charging Station
- Hospital
- Safe Flight Corridor
- Emergency Shelter

Each location represents a **state**, while the flight paths between locations represent **transitions**.

Goal Test

The goal is achieved when:

- **Current Location = Emergency Shelter**
- **Emergency Supplies Delivered = True**

Path Cost

Each movement between connected locations costs **1 unit**. The objective is to find the shortest and safest route with the minimum total path cost.

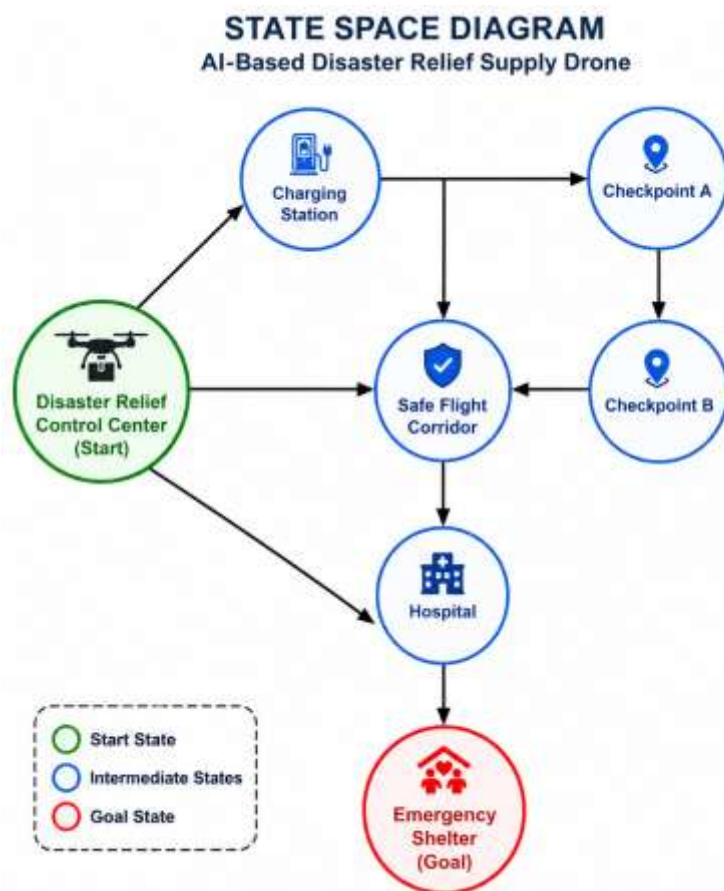
Objective

Minimize the total path cost while ensuring the safe and timely delivery of emergency supplies.

6. State Space Diagram

Explanation

- Represents the disaster area as a graph of connected locations.
- Each checkpoint, relief center, and shelter is represented as a **node**.
- Safe flight paths between locations are represented as **edges**.
- The drone starts from the **Disaster Relief Control Center (Start)**.
- The destination is the **Emergency Shelter (Goal)**.
- **Breadth-First Search (BFS)** explores each connected location level by level to determine the shortest and safest route.
- Since each flight movement has an equal cost, **BFS guarantees the optimal path** for emergency supply delivery.
- The state-space diagram enables efficient navigation, reduces delivery time, and improves the success of disaster relief missions.



7. Python Implementation

Breadth-First Search (BFS) Algorithm

```
from collections import deque

disaster_map = {
    'Relief Center': ['Charging Station', 'Safe Flight Corridor'],
    'Charging Station': ['Relief Center', 'Checkpoint A'],
    'Checkpoint A': ['Charging Station', 'Checkpoint B'],
    'Checkpoint B': ['Checkpoint A', 'Safe Flight Corridor'],
    'Safe Flight Corridor': ['Relief Center', 'Checkpoint B', 'Hospital'],
    'Hospital': ['Safe Flight Corridor', 'Emergency Shelter'],
    'Emergency Shelter': ['Hospital']
}

def bfs(graph, start, goal):
    queue = deque([[start]])
    visited = set()
    while queue:
        path = queue.popleft()
        node = path[-1]
        if node in visited:
            continue
        visited.add(node)
        if node == goal:
            return path
        for neighbour in graph[node]:
            if neighbour not in visited:
                new_path = list(path)
                new_path.append(neighbour)
                queue.append(new_path)

    return None
```



```

path = bfs(disaster_map, 'Relief Center', 'Emergency Shelter')
print("=" * 60)
print("AI-BASED DISASTER RELIEF SUPPLY DRONE")
print("=" * 60)
if path:
    print("\nOptimal Delivery Path:")
    print(" -> ".join(path))
    print("\nTotal Steps:", len(path) - 1)
    print("\nEmergency supplies delivered successfully!")
else:
    print("No valid route found.")

```

8. Output

```

=====
AI-BASED DISASTER RELIEF SUPPLY DRONE
=====

```

Optimal Delivery Path:

Relief Center

->

Safe Flight Corridor

->

Hospital

->

Emergency Shelter

Total Steps: 3

Emergency supplies delivered successfully!

9. Complexity Comparison

Algorithm	Time Complexity	Space Complexity	Path Cost	Optimal	Complete
BFS	$O(b^d)$	$O(b^d)$	Optimal	Yes	Yes
DFS	$O(b^m)$	$O(bm)$	Not Always Optimal	No	No
A*	$O(b^d)$	$O(b^d)$	Optimal	Yes	Yes

Explanation

- **BFS** is selected because it always finds the shortest path when each flight movement has an equal cost.
- **DFS** may reach the destination faster in some cases but does not always guarantee the shortest route.
- **A*** performs more efficiently when a suitable heuristic, such as the estimated distance to the destination, is available.

For this disaster relief mission:

- **Branching Factor (b) ≈ 3**
- **Solution Depth (d) = 3**

The shortest delivery route requires **3 movements**, making **Breadth-First Search (BFS)** an appropriate algorithm for autonomous disaster relief drone navigation.

10. Conclusion

The AI-Based Disaster Relief Supply Drone demonstrates how Artificial Intelligence (AI) can significantly enhance disaster response and emergency logistics. By representing the disaster area as a state-space graph and applying the Breadth-First Search (BFS) algorithm, the drone is capable of identifying the shortest and safest route to deliver essential supplies such as medicines, food, drinking water, and first-aid kits to affected communities. This intelligent navigation system minimizes delivery time, reduces operational risks, and ensures that emergency resources reach their destination efficiently, even when conventional transportation is unavailable.

The proposed system also reduces the workload and exposure of rescue personnel by performing autonomous aerial deliveries in hazardous environments. Its ability to navigate dynamically through disaster-affected regions improves the overall effectiveness of relief operations and supports faster emergency response during critical situations.

Future enhancements may include AI-based obstacle avoidance using computer vision, real-time weather monitoring, dynamic route optimization with the A* algorithm, swarm drone coordination for large-scale disaster management, battery optimization, and integration with IoT-based emergency monitoring systems. These advancements will improve the drone's autonomy, reliability, and operational efficiency, making it a valuable solution for modern disaster management and humanitarian relief missions.